Non-Homomorphism Factors and the Concentration Parameter in Graph Homomorphisms

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November 4, 2026

Abstract

The *non-homomorphism factor* $|G, H|$ is the minimum number of edges that must be removed from G so that the remaining graph admits a homomorphism to H. The *concentration parameter* $\gamma(G, H)$ is the minimum, over all homomorphisms $f\colon G \to H$, of the largest number of edges of G that f sends onto a single edge of H.

For $|G, H|$ we prove that, for the Hamming graph $H(d, n) = K_n^{\square d}$,

$$|H(d,n), H| = d\, n^{d-1}\Big(\binom{n}{2} - t\big(n, \omega(H)\big)\Big).$$

So on complete graphs, rook's graphs and all Hamming graphs only the clique number of the target matters. For every edge-transitive G and every F with $F \to G$ and $F \not\to H$ we prove $|G, H| \geq |E(G)|/|E(F)|$. An averaging argument gives the general upper bound $|V(H)|^2\,|G, H| \leq |E(G)|\,(|V(H)|^2 - 2|E(H)|)$. For a d-regular graph G on n vertices and K_{r+1}-free H we prove $|L(G), H| \geq n\big(\binom{d}{2} - t(d, r)\big)$, with equality for every H of clique number r when G has a proper d-edge-colouring. This gives exact ground-state energies on the kagome, checkerboard and pyrochlore lattices for every target. We determine the max-r-cut of the triangular graphs $J(n, 2)$ for all n and r; a parity obstruction costs exactly $r/2$ edges when n is odd and $n - 1$ is an odd multiple of r, and $J(n, 2)$ with $n \equiv 3 \pmod 4$ has semidefinite integrality gap exactly one. For odd n the folded cube satisfies $|FQ_n, H| = 2^{n-2}$ for every H with an edge into which C_n does not map. For $r = 2$ equality holds when G has a balanced 2-edge-colouring. We prove the quantitative Hedetniemi bound

$$|G \times H, K| \leq \min(2|E(H)|\,|G, K|,\, 2|E(G)|\,|H, K|)$$

and the exact value $|C_p \times C_q, H| = 2p$ for odd $p \leq q$ and every target H with an edge into which C_q does not map, for instance every odd cycle C_m with $m > q$. For cubic G we show that $|L(G), L(P)|$, where P is the Petersen graph, is at least the number of leaf blocks of the bridge-block tree of G.

For γ we prove multiplicativity, monotonicity, and the transfer bound $|G, K| \leq \gamma(G, H)\,|H, K|$. We find an exact identity for the loads of a homomorphism to K_3 in terms of the degree sums of its colour classes. From it, every d-regular graph on n vertices with an equitable 3-colouring has $\gamma(G, K_3) = \frac{d}{2}\big(\lceil n/3 \rceil + [n \equiv 2\ (3)]\big)$. With the Chen–Lih–Wu theorem this gives $\gamma(G, K_3) = 3\lceil |V(G)|/6 \rceil$ for every connected cubic graph other than K_4 and $K_{3,3}$: prisms, Möbius ladders, generalized Petersen graphs and the Petersen graph itself. For hypercubes, $\gamma(Q_d, K_3) = d(2^d + 2)/6$ for even d and $d(2^d + 4)/6$ for odd d. For odd cycles as targets we show that every d-regular G satisfies $\gamma(G, C_m) \geq \frac{d}{2} M_m(n)$, where $M_m(n)$ is the least $M \equiv n \pmod 2$ with $mM \geq n$; this is exact for cycles and tori. We also determine $\gamma(\cdot, K_3)$ for complete bipartite graphs, books, friendship graphs, wheels and complete tripartite graphs. For odd m we determine $\gamma(C_n, C_m)$ exactly: it is the least $M \equiv n \pmod 2$ with $mM \geq n$. We also show that $\max(|G, H|, |H, G|)$ is not a pseudo-metric.

Every result in this paper is formally verified in the Lean 4 proof assistant with the Mathlib library.

1 Introduction

All graphs are finite, simple and undirected.

Definition 1.1 (Graph homomorphism)**.** A *homomorphism* from G to H is a map $f\colon V(G) \to V(H)$ such that $\{u, v\} \in E(G)$ implies $\{f(u), f(v)\} \in E(H)$. We write $G \to H$ when such a map exists, and $G \leftrightarrow H$ when $G \to H$ and $H \to G$.

A proper k-colouring of G is the same as a homomorphism $G \to K_k$. Two standard facts [3] follow directly.

Proposition 1.2. *If $G \to H$ then $\chi(G) \le \chi(H)$.*

Proposition 1.3. *Every homomorphism $f\colon G \to H$ satisfies $d_H(f(u), f(v)) \le d_G(u, v)$ for all $u, v \in V(G)$.*

Proof. A walk of length ℓ from u to v maps to a walk of length ℓ from $f(u)$ to $f(v)$. □

Formal verification. Every theorem, lemma, proposition and corollary in this paper has been proved in Lean 4 [9] using Mathlib [17]. The only exceptions are the classical results quoted in Remark 2.13, the Chen–Lih–Wu theorem used in Corollary 3.15, Kőnig's theorem in Corollary 2.15, and the computer searches, which are marked as such. The development is in the `lean/NonHom` directory of the repository. Section 6 describes it and lists the Lean name of each result.

Related work. The quantity $|E(G)| - |G, H|$ is the optimum of the *maximum H-colourable subgraph* problem (Max H-Col), a restriction of Max CSP studied by Jonsson, Färnqvist, Thapper and others [11, 12]. For $H = K_k$ it is the max-k-cut. Färnqvist, Jonsson and Thapper introduced the parameter $s(H, G)$, which for edge-transitive G equals the largest H-colourable fraction of $E(G)$; our Theorem 2.10 is the corresponding averaging bound and should be read as a reformulation of their argument. Eigenvalue and semidefinite bounds for the max-k-cut go back to Mohar and Poljak [13]; van Dam and Sotirov [14] proved them tight for some graphs of the Hamming scheme with $k = q$. We found no exact max-k-cut values in the literature for the Cartesian Hamming graphs with $k < q$, for the triangular graphs $J(n, 2)$, or for tensor products of odd cycles, and no treatment of the concentration parameter γ.

2 Non-Homomorphism Factors

Definition 2.1 (Non-homomorphism factor [1])**.** $|G, H|$ is the minimum number of edges that must be removed from G to obtain a graph G' with $G' \to H$. In particular $|G, H| = 0$ if and only if $G \to H$.

Remark 2.2 (Complexity)**.** For fixed H, deciding $|G, H| \le k$ is Edge Deletion-H-Colouring, studied by Foucaud, Hocquard, Lajou, Mitsou and Pierron [4]. The case $k = 0$ is H-Colouring. So by the Hell–Nešetřil dichotomy [3], computing $|G, H|$ is NP-hard whenever the core of H has more than one edge. For $H = K_2$ the problem is Edge Bipartization, which is NP-complete but fixed-parameter tractable in k.

Proposition 2.3 ([1])**.** *1. If $H_1 \to H_2$ then $|G, H_1| \ge |G, H_2|$ for every G.*

2. If $H_1 \leftrightarrow H_2$ then $|G, H_1| = |G, H_2|$ for every G.

3. If $f\colon G_1 \to G_2$ is edge-injective (distinct edges map to distinct edges) then $|G_1, H| \le |G_2, H|$ for every H.

Proof. (1) If $G' \to H_1$ then $G' \to H_1 \to H_2$. (2) Apply (1) in both directions. (3) Take $S \subseteq E(G_2)$ with $|S| = |G_2, H|$ and $G_2 - S \to H$, and let $S' = \{e \in E(G_1) : f(e) \in S\}$. Edge-injectivity gives $|S'| \le |S|$, and f restricts to a homomorphism $G_1 - S' \to G_2 - S$. □

2.1 Complete graphs and Turán's theorem

Let $t(n, m)$ be the number of edges of the Turán graph $T(n, m)$, the complete m-partite graph on n vertices with parts as equal as possible. By Turán's theorem, every K_{m+1}-free graph on n vertices has at most $t(n, m)$ edges.

Theorem 2.4. *For $m \ge 1$, $|K_n, K_m| = \binom{n}{2} - t(n, m)$. If $n = qm + r$ with $0 \le r < m$, this equals $r\binom{q+1}{2} + (m - r)\binom{q}{2}$. In particular $|K_{n+1}, K_n| = 1$ for all $n \ge 1$.*

Proof. Removing the edges outside $T(n, m)$ leaves an m-colourable graph. Conversely, an m-colourable spanning subgraph of K_n is K_{m+1}-free, so by Turán's theorem it has at most $t(n, m)$ edges. The explicit formula follows by induction on q from $t(n+m, m) = t(n, m) + n(m-1) + \binom{m}{2}$. For $n + 1 = 1 \cdot n + 1$ it gives $\binom{2}{2} = 1$. □

Table 1: $|K_n, K_m|$ for $n = 2, \dots, 7$ and $m = 2, \dots, 6$, from Theorem 2.4.

	K_2	K_3	K_4	K_5	K_6
K_2	0	0	0	0	0
K_3	1	0	0	0	0
K_4	2	1	0	0	0
K_5	4	2	1	0	0
K_6	6	3	2	1	0
K_7	9	5	3	2	1

Theorem 2.5. *For all graphs G and H, $|G, H| \geq \max\big(0,\, |E(G)| - t(|V(G)|, \chi(H))\big)$.*

Proof. If $G' \subseteq G$ and $G' \to H$ then $\chi(G') \leq \chi(H)$, so G' is $K_{\chi(H)+1}$-free and has at most $t(|V(G)|, \chi(H))$ edges. □

For complete sources, the target matters only through its clique number.

Theorem 2.6. *Let H have clique number $r \geq 1$. Then $|K_n, H| = \binom{n}{2} - t(n, r)$. In particular $|K_n, H| = \binom{n}{2} - \lfloor n^2/4 \rfloor$ for every triangle-free H with an edge.*

Proof. A homomorphism maps a clique injectively onto a clique. So if $G' \to H$ then G' is K_{r+1}-free, and by Turán's theorem $|E(G')| \leq t(n, r)$. Conversely $T(n, r) \to K_r \to H$. □

Theorem 2.4 is the case $H = K_m$. For example $|K_n, C_{2k+1}|$ and $|K_n, P|$ do not depend on k or on the Petersen structure. They all equal $|K_n, K_2|$, while $|K_n, K_3|$ is strictly smaller for $n \geq 3$.

The same holds for every Hamming graph. The *Hamming graph* $H(d, n)$ has vertex set $\{0, \dots, n-1\}^d$, two words being adjacent if they differ in exactly one coordinate. So $H(1, n) = K_n$, $H(2, n) = K_n \square K_n$ is the $n \times n$ rook's graph, and $H(d, 2) = Q_d$ is the hypercube.

Theorem 2.7. *Let H have clique number $r \geq 1$. Then*

$$|H(d, n), H| = d\, n^{d-1} \left(\binom{n}{2} - t(n, r) \right).$$

Proof. A *line* is the set of n words that agree outside one fixed coordinate. Every line induces K_n, and every edge lies on exactly one line, so $E(H(d, n))$ is partitioned into the edge sets of $d\, n^{d-1}$ copies of K_n. If $H(d, n) - S \to H$, the kept part of each line is K_{r+1}-free. By Turán's theorem each line therefore loses at least $\binom{n}{2} - t(n, r)$ edges.

For the upper bound, colour a word x by $\big(\sum_i x_i \bmod n\big) \bmod r$. This is a homomorphism to K_r, hence to H, after deleting the monochromatic edges. Along a line the value $\sum_i x_i \bmod n$ runs through all residues exactly once. So the colouring of each line is a relabelled Turán graph $T(n, r)$, and exactly $\binom{n}{2} - t(n, r)$ of its edges are monochromatic. □

For example, the $n \times n$ rook's graph has $|K_n \square K_n, K_2| = 2n\big(\binom{n}{2} - \lfloor n^2/4 \rfloor\big)$. Every triangle-free target with an edge, such as C_5 or P, gives the same value, while $|K_n \square K_n, K_3| = 2n\big(\binom{n}{2} - t(n, 3)\big)$. Theorem 2.6 is the case $d = 1$.

The density bound of Theorem 2.5 has a matching upper bound. It comes from a uniformly random map.

Theorem 2.8 (Averaging bound)**.** *For all graphs G and every graph H with at least one vertex,*

$$|V(H)|^2\, |G, H| \;\leq\; |E(G)|\, \big(|V(H)|^2 - 2|E(H)|\big).$$

In particular $|G, K_m| \leq |E(G)|/m$.

Proof. For a map $f \colon V(G) \to V(H)$, delete the edges whose image is not an edge. An edge uv of G survives for exactly $2|E(H)| \cdot |V(H)|^{|V(G)|-2}$ maps f. So the average number of deleted edges is $|E(G)|(1 - 2|E(H)|/|V(H)|^2)$, and some map does at least this well. □

2.2 Cycles and the Petersen graph

Every cycle is 3-colourable, so $|C_n, K_3| = 0$. For K_2, $|C_n, K_2| = 0$ if n is even and $|C_n, K_2| = 1$ if $n \geq 3$ is odd. Indeed an odd cycle is not bipartite, and deleting one edge leaves a path.

Let P be the Petersen graph, with outer cycle $u_0u_1u_2u_3u_4$, spokes u_iv_i, and inner edges v_iv_{i+2} (indices mod 5). It has 10 vertices, 15 edges, and $\chi(P) = 3$, so $|P, K_m| = 0$ for $m \geq 3$.

Theorem 2.9. $|P, K_2| = 3$.

Proof. P has exactly twelve 5-cycles, and each edge lies on exactly four of them. Two edges therefore lie on at most eight, so after deleting any two edges some 5-cycle survives and the graph is not bipartite. Hence $|P, K_2| \geq 3$. Conversely, deleting u_0u_1, u_3v_3 and v_2v_4 leaves a bipartite graph with sides $\{u_0, u_1, u_3, v_2, v_3, v_4\}$ and $\{u_2, u_4, v_0, v_1\}$. □

Theorem 2.5 only gives $|P, K_2| \geq 15 - t(10, 2) < 0$ here, so the odd-cycle structure, not edge density, is what forces the deletions.

2.3 Edge-transitive graphs

A graph G is *edge-transitive* if its automorphism group acts transitively on $E(G)$.

Theorem 2.10. *Let G be edge-transitive. Let F be a graph with at least one edge such that $F \to G$ and $F \not\to H$. Then*

$$|G, H| \;\geq\; \frac{|E(G)|}{|E(F)|}.$$

Proof. Fix $\varphi\colon F \to G$ and let $\Gamma = \mathrm{Aut}(G)$. For an edge e of G, put $c(e) = \sum_{\sigma\in\Gamma} \mu(\sigma \circ \varphi, e)$. Reindexing by $\sigma \mapsto \tau^{-1}\sigma$ gives $c(\tau e) = c(e)$. So by edge-transitivity c is a constant c_0, and summing over all edges gives $|E(G)|\, c_0 = |\Gamma|\, |E(F)|$.

Now let $G - S \to H$. Since $F \not\to H$, each $\sigma\circ\varphi$ maps some edge of F into S, so $|\Gamma| \leq \sum_{e\in S} c(e) = |S|\, c_0$. Hence $|E(G)| \leq |E(F)|\, |S|$. □

With F an odd cycle and $H = K_2$, this recovers $|P, K_2| \geq 15/5 = 3$ (Theorem 2.9). It also recovers $|C_p \times C_q, K_2| \geq 2pq/q = 2p$ (Theorem 2.22), since $C_p \times C_q$ is edge-transitive and contains closed odd walks of length q. Similarly, for an edge-transitive G containing K_{r+1} and a K_{r+1}-free H, the theorem gives $|G, H| \geq |E(G)|/\binom{r+1}{2}$. Together with Theorem 2.8, the parameter $|G, H|$ of an edge-transitive graph is pinned between two linear functions of $|E(G)|$.

The bound is sharp on the folded cubes. The *folded cube* FQ_n has vertex set $\{0, 1\}^{n-1}$, two words being adjacent if they differ in exactly one coordinate or in all coordinates. It is the quotient of Q_n by antipodal pairs, has 2^{n-1} vertices and $n2^{n-2}$ edges, and for odd n its shortest odd cycle has length n.

Theorem 2.11. *Let $n \geq 3$ be odd, and let H have an edge with $C_n \not\to H$. Then*

$$|FQ_n, H| = 2^{n-2}.$$

In particular $|FQ_n, K_2| = 2^{n-2}$ and $|FQ_n, C_m| = 2^{n-2}$ for every odd $m > n$.

Proof. For each vertex x, flip the coordinates $1, \ldots, n-1$ one at a time and then complement. This is a closed walk of length n, so a homomorphism $C_n \to FQ_n$, and it must use a deleted edge. An edge determines the step at which a walk uses it, by the set of coordinates its ends differ in, and then x up to the choice of endpoint. So each edge lies on at most two of the 2^{n-1} walks, and $|S| \geq 2^{n-2}$. Conversely, deleting the 2^{n-2} antipodal edges leaves Q_{n-1}, which is bipartite, and $K_2 \to H$. □

2.4 Line graphs of regular graphs

Theorem 2.12. *Let G be d-regular on n vertices and let H be K_{r+1}-free. Then*

$$|L(G), H| \;\geq\; n\left(\binom{d}{2} - t(d, r)\right).$$

For $r = 2$ this reads $|L(G), H| \geq n\lfloor(d-1)^2/4\rfloor$. Suppose moreover that H has an edge and that $E(G)$ has a 2-colouring that is balanced*, meaning that at every vertex the two colours occur $\lfloor d/2\rfloor$ and $\lceil d/2\rceil$ times. Then equality holds for every triangle-free H.*

Proof. The d edges at a vertex v form a clique $Q_v \cong K_d$ in $L(G)$. Distinct cliques share no edge of $L(G)$, since two distinct edges of G share at most one vertex. If $L(G) - S \to H$ then $L(G) - S$ is K_{r+1}-free. By Turán's theorem it keeps at most $t(d,r)$ of the $\binom{d}{2}$ edges of each Q_v. Summing over v gives the lower bound.

For the upper bound, colour the vertices of $L(G)$ by the given edge colouring, and delete the edges of $L(G)$ whose ends have the same colour. What remains maps to $K_2 \to H$. At a vertex with colour counts a and $d-a$ the deleted edges number $\binom{a}{2} + \binom{d-a}{2} = \binom{d}{2} - a(d-a)$. This equals $\binom{d}{2} - \lfloor d^2/4 \rfloor$ when the colouring is balanced. □

Remark 2.13. Balanced 2-edge-colourings exist in three cases. If $4 \mid d$, split a 2-factorization (Petersen's 2-factor theorem [6]) evenly. If G is bipartite, group the d colour classes of a proper edge colouring (Kőnig [7]). If d is odd and G has a perfect matching M, split $G - M$ into $(d-1)/2$ two-factors. Then give colour 1 to M together with $\lfloor (d-1)/4 \rfloor$ of these two-factors.

In particular every bridgeless cubic graph G on n vertices has a perfect matching, by Petersen's theorem [6]. So $|L(G), H| = n$ for every triangle-free H with an edge, and the whole ladder $|L(G), C_5| = |L(G), C_7| = \cdots = |L(G), K_2|$ collapses.

These two classical theorems are not formalized. The Lean proof establishes the cubic case for every cubic graph that has a perfect matching.

For graphs with a proper d-edge-colouring the lower bound of Theorem 2.12 is exact for every target, not only for $r = 2$.

Theorem 2.14. *Let G be d-regular on n vertices with a proper d-edge-colouring, and let H have clique number $r \geq 1$. Then*

$$|L(G), H| = n \left(\binom{d}{2} - t(d,r) \right).$$

Proof. The lower bound is Theorem 2.12. For the upper bound, give each edge of G its colour modulo r and map $L(G)$ to $K_r \subseteq H$, deleting the monochromatic pairs. At a vertex v the d incident edges carry the d colours once each, so the induced colouring of the clique Q_v is a relabelled Turán graph $T(d,r)$, and exactly $\binom{d}{2} - t(d,r)$ pairs at v are deleted. □

By Kőnig's theorem [7] every bipartite regular graph has such a colouring. This gives exact ground-state energies of the antiferromagnetic q-state Potts model, and of every constrained spin model with target H, on the corner-sharing lattices.

Corollary 2.15. *Let G be a finite bipartite d-regular graph on n vertices, for example a honeycomb, square or diamond lattice on a torus with even periods. For every H with clique number r,*

$$|L(G), H| = n \left(\binom{d}{2} - t(d,r) \right).$$

For the kagome lattice $L(\text{honeycomb})$ this is 1 unsatisfied bond per triangle for $r = 2$ and 0 for $r \geq 3$. For the pyrochlore lattice $L(\text{diamond})$ and the checkerboard lattice $L(\text{square})$ it is 2 per tetrahedron, respectively crossed square, for $r = 2$, 1 for $r = 3$ and 0 for $r \geq 4$.

The Ising values are the classical counts of one frustrated bond per kagome triangle and of the ice rule on pyrochlore. What is new is the exact value for every target at once. In particular the 3-state Potts antiferromagnet on pyrochlore has exactly one unsatisfied bond per tetrahedron.

Theorem 2.16. *Let C_n^2 ($n \geq 5$) be the circulant graph on $\mathbb{Z}_n$ with jumps 1 and 2. For every triangle-free H, $|C_n^2, H| \geq \lceil n/2 \rceil$.*

Proof. The n triangles $\{i, i+1, i+2\}$ must each lose an edge. An edge $\{i, i+1\}$ lies only in the triangles starting at $i-1$ and at i, and an edge $\{i, i+2\}$ lies only in the triangle starting at i. So each deleted edge destroys at most two of these triangles. □

2.5 Triangular graphs

The triangular graph $L(K_n)$ is $(2n-4)$-regular, and Theorem 2.12 gives $|L(K_n), K_2| \geq n\lfloor (n-2)^2/4 \rfloor$. Parity decides when this is exact.

Theorem 2.17. *For $n \geq 2$,*

$$|L(K_n), K_2| = n \left\lfloor \frac{(n-2)^2}{4} \right\rfloor + [\, n \equiv 3 \pmod 4 \,].$$

Proof. A map $L(K_n)-S \to K_2$ is a 2-colouring of $E(K_n)$. If vertex v has a_v edges of colour 0, then S must contain the $\binom{n-1}{2} - a_v(n-1-a_v) \geq \lfloor (n-2)^2/4 \rfloor$ monochromatic pairs at v. Conversely, deleting exactly the monochromatic pairs suffices. Equality at v needs a_v to be balanced. For $n \equiv 3 \pmod 4$ this means $a_v = (n-1)/2$, which is odd, at an odd number n of vertices. That contradicts the handshake lemma, so some vertex costs at least one more.

For the upper bounds: if n is even, colour uw by the parity of $u+w$. If $n = 2m+1$, give colour 0 to the pairs at circular distance at most $\lceil m/2 \rceil$. When m is odd, remove the near-perfect matching $\{0,1\},\{2,3\},\dots,\{2m-2,2m-1\}$ from colour 0. Then every vertex is balanced except the vertex $2m$. □

So $|L(K_n), K_2| = 1, 4, 10, 24, 43, 72, 108$ for $n = 3, \dots, 9$. For $n = 7$ an exhaustive search shows that no triangle-free target at all reaches the line-graph bound 42. Every $K_{3,3}$-balanced choice of kept edges at the seven vertices contains a triangle. Hence $|L(K_7), H| = 43$ for every triangle-free H with an edge; this search is not formalized.

A SAT computation extends this to $n = 11$. Call a vertex u of a triple T a separator if the local bipartition at u splits the other two vertices of T. Kept edges reaching the bound form a triangle-free graph exactly when no triple has three separators. For balanced local bipartitions, a parity argument shows that having no triple with one or three separators is impossible when $n \equiv 3 \pmod 4$; this is the handshake argument above. The solver CaDiCaL shows that, for odd $n \leq 13$, no triple with three separators forces no triple with one separator. For $n = 11$ this takes 0.2 seconds, and two other solvers agree. Hence $|L(K_{11}), H| = |L(K_{11}), K_2| = 221$ for every triangle-free H with an edge. Simulated annealing never found fewer than one fully separated triple for $n = 15$, 19 and 23.

For $r \geq 3$ the excess is a property of the target K_r, not of clique number r.

Theorem 2.18. *Let $n \geq 1$ and $r \geq 3$. Then there is a graph K of clique number r with*

$$|L(K_n), K| = n\left(\binom{n-1}{2} - t(n-1,r)\right).$$

So the minimum of $|L(K_n), H|$ over all H of clique number r is the Turán value, also in the exceptional case of Theorem 2.19.

Proof. At every vertex v of K_n choose an equitable r-colouring c_v of the other $n-1$ vertices. Let K be the graph on $E(K_n)$ in which $vu \sim vw$ when $c_v(u) \neq c_v(w)$. The identity map sends $L(K_n)$ minus the monochromatic pairs to K, and there are $n\big(\binom{n-1}{2} - t(n-1,r)\big)$ such pairs. Four or more edges of K_n that pairwise meet form a star, so every clique of K with more than three vertices lies in one star, where K is complete r-partite. Since $r \geq 3$, K has clique number r. The lower bound is Theorem 2.12. □

The same parity phenomenon governs every number of colours.

Theorem 2.19. *Let $n \geq 3$, $r \geq 2$, and write $n-1 = qr+s$ with $0 \leq s < r$. Then*

$$|L(K_n), K_r| = n\left(\binom{n-1}{2} - t(n-1,r)\right) + \frac{r}{2}\left[n \text{ odd},\ s = 0,\ q \text{ odd}\right].$$

Equivalently, the max-r-cut of the triangular graph $J(n,2)$ is $n\,t(n-1,r)$, minus $r/2$ in the exceptional case. For even n the value $n\big(\binom{n-1}{2} - t(n-1,r)\big)$ holds for every target of clique number r. Outside the exceptional case the same is true for odd n.

Proof. A map $L(K_n) - S \to K_r$ is an r-colouring of $E(K_n)$, and S contains all monochromatic pairs at each vertex: $\sum_c \binom{d_c(v)}{2}$ of them at v, where $d_c(v)$ is the number of edges of colour c at v. This is at least $\binom{n-1}{2} - t(n-1,r)$, with equality exactly when the $d_c(v)$ are equitable.

In the exceptional case every $d_c(v)$ would have to equal q, but then colour class c would have odd degree sum nq. Writing $x_c(v) = d_c(v) - q$, the excess at v is $\frac{1}{2}\sum_c x_c(v)^2$, and every colour c needs some v with $x_c(v)$ odd. So the total excess is at least $r/2$.

For the constructions:

- If n is even, colour a 1-factorization of K_n modulo r and apply Theorem 2.14.
- If $r \nmid n-1$, write $n-1 = kr+s$ with $0 < s < r$, put $B = \{0,\dots,kr-1\}$ and $A = \{kr,\dots,n-1\}$, and let $\theta(x) = \lfloor x/2 \rfloor + (x \bmod 2)\lceil r/2 \rceil$, a permutation of $\mathbb{Z}_r$ with $2\theta(x) \in \{x, x-1\}$. Colour uw by $\theta(u)+\theta(w) \bmod r$ inside B, by $(w-kr)+u-1 \bmod r$ for $u \in B$, $w \in A$, and by $(u-kr)+(w-kr) \bmod (s+1)$ inside A. Every colour then occurs k or $k+1$ times at every vertex. This explicit colouring replaces the existence theorem of Hilton and de Werra [15].

- If $n = 2m+1$ is odd and $s = 0$, use the centre classes: the edges uw with $u+w \equiv 2h \pmod n$ form a near-perfect matching missing h. Give each colour q of the classes $h \neq 0$, choosing the colours of h and $-h$ equal for as many h as possible, and colour an edge $\{u, -u\}$ of class 0 like the class of its smaller endpoint. A vertex then sees every colour q times, except that a pair $h, -h$ of differently coloured classes costs one unit. If q is even no such pair is needed. If q is odd, exactly $r/2$ pairs are needed. This replaces the Walecki decomposition [16].

□

For $r = 2$ this is Theorem 2.17. For $r = n-1$ and odd n it gives $|L(K_n), K_{n-1}| = (n-1)/2$, so K_n falls short of an $(n-1)$-edge-colouring by exactly $(n-1)/2$ monochromatic pairs.

Corollary 2.20. *The Mohar–Poljak eigenvalue bound $\frac{1}{4}|V|\lambda_{\max}(L)$ on the max-cut of $J(n,2)$ equals $\frac{1}{4}n(n-1)^2$. It is attained exactly when $n \equiv 1 \pmod 4$, it exceeds the max-cut by 1 when $n \equiv 3 \pmod 4$, and by $n/4$ when n is even. For the Hamming graph $H(d,q)$ the eigenvalue bound $\frac{k-1}{2k}|V|\lambda_{\max}(L)$ on the max-k-cut is attained exactly when $k \mid q$, and otherwise exceeds it by $d\,q^{d-1}e(k-e)/(2k)$, where $e = q \bmod k$.*

Proof. For a k-colouring c with indicator vectors $x_1, \dots, x_k$, the cut size is $\frac{1}{2}\sum_i x_i^{\mathsf T} L x_i$. If $x^{\mathsf T}Lx \le \lambda\|x\|^2$ for all x, applying this to $x_i - \frac{1}{k}\mathbf{1}$ gives the bound $\frac{k-1}{2k}\lambda|V|$. A cover of the edges by cliques C_j, each vertex in ρ cliques of size s, gives $x^{\mathsf T}Lx = \rho s\|x\|^2 - \sum_j(\sum_{C_j} x)^2 \le \rho s\|x\|^2$. The stars of K_n cover $J(n,2)$ with $\rho = 2$, $s = n-1$; the lines cover $H(d,q)$ with $\rho = d$, $s = q$. So $\lambda = 2n-2$ and $\lambda = dq$ are valid. A vector supported on a 4-cycle, or on two coordinates, shows that no smaller λ is valid. Compare with Theorems 2.17 and 2.7, using $t(q,k) \le \frac{k-1}{2k}q^2$ with equality iff $k \mid q$. □

Both graphs are walk-regular, so for $k = 2$ the eigenvalue bound coincides with the Goemans–Williamson semidefinite bound [14]. So $J(n,2)$ with $n \equiv 3 \pmod 4$ is an explicit family whose semidefinite integrality gap is exactly one edge. The proof above uses no eigenvalue theory, and it is formalized as stated.

2.6 Tensor products

The tensor product $G \times H$ has vertex set $V(G) \times V(H)$, with $(g,h) \sim (g',h')$ if and only if $g \sim g'$ and $h \sim h'$.

Theorem 2.21 (Quantitative Hedetniemi bound)**.** *For all graphs G, H and every graph K with at least one vertex,*

$$|G \times H, K| \;\le\; \min\bigl(2|E(H)| \cdot |G,K|,\; 2|E(G)| \cdot |H,K|\bigr).$$

Proof. The projection $G \times H \to G$ sends exactly $2|E(H)|$ edges onto each edge of G, one for each orientation of each edge of H. Now apply Lemma 3.5 to this map. If $G \times H$ has no edges it maps to any graph with a vertex. The second bound is symmetric. □

The hypothesis on K is needed. If K has no vertices and H has no edges, the right-hand side is 0 but the left-hand side is infinite. By Shitov's counterexample to Hedetniemi's conjecture [5], the bound can be far from tight for large targets: the left-hand side can be 0 while neither factor is k-colourable. For two odd cycles it is tight, and not only for $K = K_2$.

Theorem 2.22. *Let $3 \le p \le q$ be odd and let H have an edge. Then*

$$|C_p \times C_q, H| = \begin{cases} 2p & \text{if } C_q \not\to H, \\ 0 & \text{if } C_q \to H. \end{cases}$$

In particular $|C_p \times C_q, C_m| = 2p$ for every odd $m > q$.

Proof. If $C_q \to H$, compose with the projection $C_p \times C_q \to C_q$. Otherwise $|C_p \times C_q, H| \le |C_p \times C_q, K_2|$ by Proposition 2.3, since $K_2 \to H$. The right-hand side is at most $2p$ by Theorem 2.21, since $|C_q, K_2| = 1$.

For the lower bound, let $\varepsilon_i = +1$ for $0 \le i < (q+p)/2$ and $\varepsilon_i = -1$ for $(q+p)/2 \le i < q$, and let $g_i = \sum_{j<i}\varepsilon_j$. Then $g_q = p \equiv 0 \pmod p$. For $a \in \mathbb{Z}_p$, $h \in \mathbb{Z}_q$ and $\sigma \in \{\pm 1\}$, the walk $i \mapsto (a + \sigma g_i,\, h+i)$, $0 \le i \le q$, is a closed walk of length q in $C_p \times C_q$, that is, a homomorphism $C_q \to C_p \times C_q$. If one of these $2pq$ walks avoided the deletion set S, composing with $C_p \times C_q - S \to H$ would give $C_q \to H$. So each walk contains a deleted edge.

Fix a step index i and an edge e. The second coordinate increases by 1 along every step, so it determines which end of e the step leaves from. The first coordinate then determines $\sigma\varepsilon_i$, hence σ, and then a and h. So at most one walk uses e at step i. Charging each walk to the step at which it meets the deletion set S gives $2pq \le q|S|$. □

2.7 Petersen colourings

A *Petersen colouring* of a cubic graph G is a map $E(G) \to E(P)$ that sends the three edges at each vertex onto the three edges at some vertex of P. Since P is triangle-free, every triangle of $L(P)$ is the star of a vertex. So Petersen colourings of G are exactly the homomorphisms $L(G) \to L(P)$, and Jaeger's Petersen colouring conjecture [8] says that $|L(G), L(P)| = 0$ for every bridgeless cubic G.

Theorem 2.23. *Let G be cubic. Let $X_1, \dots, X_t \subseteq V(G)$ be pairwise disjoint sets, each of which is left by exactly one edge of G. Then $|L(G), L(P)| \geq t$. In particular $|L(G), L(P)|$ is at least the number of leaves of the bridge-block tree of G.*

Proof. Suppose $\varphi\colon L(G) - S \to L(P)$. Call $x \in V(G)$ *good* if none of the three edges of $L(G)$ between edges at x lies in S. For a good x, the images of the three edges at x form a triangle of $L(P)$, which is a star.

P has exactly six perfect matchings, and each edge lies in exactly two of them. Fix $X = X_i$, its leaving edge b, and a perfect matching M. Each good x has exactly one incident edge whose image is in M. If every vertex of X is good, counting incidences gives $|X| = 2m + [\varphi(b) \in M]$, where m is the number of edges inside X with image in M. But $3|X| = 2|E(X)| + 1$, so $|X|$ is odd and $\varphi(b) \in M$. This holds for all six matchings, which is impossible.

So each X_i contains a bad vertex x_i, together with an edge $\{e_i, f_i\} \in S$ of $L(G)$ where e_i, f_i are distinct edges at x_i. Two distinct edges of G share at most one vertex, so these t elements of S are distinct. The vertex sets of the leaf blocks of the bridge-block tree satisfy the hypothesis, each being left only by its bridge. □

3 The Concentration Parameter

Definition 3.1 (Concentration parameter)**.** For a homomorphism $f\colon G \to H$ and $e \in E(H)$, the *load* $\mu(f, e)$ is the number of edges of G mapped onto e. Let $\mu(f) = \max_e \mu(f, e)$ and

$$\gamma(G, H) = \min_{f\colon G \to H} \mu(f),$$

with $\gamma(G, H) = +\infty$ when $G \not\to H$.

A related per-homomorphism quantity M^σ appears in [2].

Proposition 3.2 (Pigeonhole bound)**.** *If $G \to H$ then $|E(G)| \leq |E(H)| \cdot \gamma(G, H)$, so $\gamma(G, H) \geq \lceil |E(G)|/|E(H)| \rceil$.*

Proof. The loads of f sum to $|E(G)|$. □

Lemma 3.3. *If $|E(G)| \geq 1$ and $G \to H$ then $\gamma(G, H) \geq 1$. Moreover $\gamma(G, G) = 1$.*

Lemma 3.4 (Multiplicativity)**.** *If $G \to H \to K$ then $\gamma(G, K) \leq \gamma(G, H) \cdot \gamma(H, K)$.*

Proof. Take optimal $f\colon G \to H$ and $g\colon H \to K$. An edge e of K has at most $\mu(g)$ preimage edges in H, and each of these has at most $\mu(f)$ preimages in G. So $\mu(g \circ f) \leq \mu(f)\mu(g)$. □

Lemma 3.5 (Transfer bound)**.** *If $G \to H$ and $|E(G)| \geq 1$, then $|G, K| \leq \gamma(G, H) \cdot |H, K|$ for every K. More generally $|G, K| \leq \mu(f)\,|H, K|$ for every homomorphism $f\colon G \to H$.*

Proof. Let $H - S \to K$ with $|S| = |H, K|$, and let T be the set of edges of G that f maps into S. Then $|T| \leq \mu(f)|S|$, and f restricts to $G - T \to H - S \to K$. □

Corollary 3.6 (Two-sided bound)**.** *If $G \to H$ and $K \to H$, and both have an edge, then* $\max(|G, K|, |K, G|) \leq \max\big(\gamma(G, H)|H, K|,\ \gamma(K, H)|H, G|\big)$.

Lemma 3.7 (Monotonicity in the target)**.** *If H_1 embeds injectively in H_2 (for instance $H_1 \subseteq H_2$), then $\gamma(G, H_2) \leq \gamma(G, H_1)$.*

Proof. Composing with an injective homomorphism does not change any load. □

Proposition 3.8. *For $2 \leq n \leq m$, $\gamma(K_n, K_m) = 1$. For $n > m$, $\gamma(K_n, K_m) = +\infty$.*

3.1 An exact identity for homomorphisms to K_3

For $f\colon G \to K_3$ with colour classes A_0, A_1, A_2, let $D_i = \sum_{v \in A_i} \deg v$ and write $L_{ij} = \mu(f, \{i, j\})$.

Lemma 3.9. *For every graph G, every $f\colon G \to K_3$ and distinct colours a, b, c,*

$$2L_{ab} = D_a + D_b - D_c, \qquad \textit{hence} \qquad 2\mu(f) = \max(D_0 + D_1 - D_2,\ D_0 + D_2 - D_1,\ D_1 + D_2 - D_0).$$

Proof. Each edge at a vertex of A_a goes to A_b or to A_c, so $D_a = L_{ab} + L_{ac}$. Solve the three equations. □

Theorem 3.10 (d-regularity formula)**.** *Let G be d-regular on n vertices with $d \geq 1$, and let $f\colon G \to K_3$ have colour classes of sizes a, b, c. Then*

$$2\mu(f) = d \cdot \max(a + b - c,\ a + c - b,\ b + c - a).$$

The triple $(a+b-c, a+c-b, b+c-a)$ is non-negative, sums to n, and has constant parity. Hence, if M is the least maximum of such a triple, then $\gamma(G, K_3) \geq \frac{d}{2} M$. Equality holds as soon as some proper 3-colouring attains M.

Proof. Put $D_i = d|A_i|$ in Lemma 3.9. Non-negativity follows from $d\,(a + b - c) = 2L_{ab} \geq 0$, and the entries differ by even numbers. □

Theorem 3.11. *For all $n \geq 3$,*

$$\gamma(C_n, K_3) = \begin{cases} \lceil n/3 \rceil & \text{if } n \not\equiv 2 \pmod 3, \\ \lceil n/3 \rceil + 1 & \text{if } n \equiv 2 \pmod 3. \end{cases}$$

In particular $\gamma(C_n, K_3)$ attains the pigeonhole bound if and only if $n \not\equiv 2 \pmod 3$.

Proof. Apply Theorem 3.10 with $d = 2$ and write $n = 3q + r$. A same-parity triple summing to n has maximum at least $\lceil n/3 \rceil$. For $r = 2$ the only triple with maximum $q + 1$ is $(q + 1, q + 1, q)$, which has mixed parity, so the maximum is at least $q + 2$.

To see that these bounds are attained, colour vertex $i < 3q$ by $i \bmod 3$. If $r = 1$, give the last vertex colour 1. If $r = 2$, give the last two vertices colours 0 and 1. The class sizes are (q, q, q), $(q, q + 1, q)$ and $(q + 1, q + 1, q)$, with maxima q, $q + 1$ and $q + 2$. □

Theorem 3.12. $|K_{n,n}, K_3| = 0$ *and* $\gamma(K_{n,n}, K_3) = n\lceil n/2 \rceil$.

Proof. Let $f\colon K_{n,n} \to K_3$. The colour sets of the two sides are disjoint, so one side is monochromatic in a colour that the other side does not use. That colour class has exactly n vertices. With $d = n$ and class sizes $(x, n - x, n)$, Theorem 3.10 gives $\mu(f) = n \max(x, n - x) \geq n\lceil n/2 \rceil$. Splitting one side as evenly as possible attains this. □

For $n = 1, 2, 3, 4$ this gives $1, 2, 6, 8$.

Theorem 3.13. $\gamma(P, K_3) = 6$ *and* $\gamma(P, K_5) = 2$.

Proof. For K_3 take $d = 3$ and $n = 10$. Same-parity triples summing to 10 are all even, so their maximum is at least 4, and $\gamma(P, K_3) \geq 6$. Equality holds because P has a proper 3-colouring with class sizes $(3, 4, 3)$, whose triple is $(4, 2, 4)$. For K_5, the pigeonhole bound $\lceil 15/10 \rceil = 2$ is attained by an explicit 5-colouring. □

So $\gamma(P, K_3) = 6$ exceeds the pigeonhole bound 5 because of parity. In contrast $\gamma(P, K_5)$ attains it.

3.2 Equitable colourings and regular graphs

A proper colouring is *equitable* if its class sizes differ pairwise by at most one. Write $M(n) = \lceil n/3 \rceil + [n \equiv 2 \pmod 3]$. This is the least maximum of a same-parity triple of non-negative integers summing to n, and it is the value found in Theorem 3.11.

Theorem 3.14. *Let G be d-regular on n vertices, $d \geq 1$, with an equitable proper 3-colouring. Then*

$$\gamma(G, K_3) = \frac{d}{2}\, M(n),$$

and every equitable 3-colouring is optimal.

Proof. The lower bound is Theorem 3.10 with $M = M(n)$. If the class sizes differ by at most one, the triple $(a+b-c, a+c-b, b+c-a)$ is (q,q,q), $(q+1,q+1,q-1)$ or $(q+2,q,q)$ for $n = 3q$, $3q+1$, $3q+2$. Its maximum is $M(n)$, so the identity of Theorem 3.10 gives $2\mu(f) = d\,M(n)$. □

Chen, Lih and Wu [10] proved, as a case of their equitable Δ-colouring theorem, that every connected cubic graph other than K_4 and $K_{3,3}$ has an equitable 3-colouring.

Corollary 3.15. *Every connected cubic graph G other than K_4 and $K_{3,3}$ satisfies*

$$\gamma(G, K_3) = 3\left\lceil \frac{|V(G)|}{6} \right\rceil .$$

Moreover $\gamma(K_{3,3}, K_3) = 6$ and $\gamma(K_4, K_3) = \infty$.

Proof. For $n = |V(G)|$ even, $\frac{3}{2}M(n) = 3\lceil n/6 \rceil$. Apply Theorem 3.14. The value for $K_{3,3}$ is Theorem 3.12. □

In particular every prism $C_k \square K_2$ ($k \geq 3$), every Möbius ladder on $2k \geq 8$ vertices, every generalized Petersen graph and the Petersen graph have $\gamma(G, K_3) = 3\lceil |V(G)|/6 \rceil$. By contrast, the pigeonhole bound is only $|V(G)|/2$. A computer check of all 45 982 connected cubic graphs on at most 18 vertices confirms that K_4 and $K_{3,3}$ are the only exceptions.

Theorem 3.16. *For $d \geq 1$,*

$$\gamma(Q_d, K_3) = \begin{cases} d\,(2^d+2)/6 & d \text{ even}, \\ d\,(2^d+4)/6 & d \text{ odd}. \end{cases}$$

Proof. Colour $x \in \{0,1\}^d$ by its weight mod 3. Adjacent words differ in weight by one, so this is proper. Let $N_c(d)$ count the words of weight $\equiv c$. These counts satisfy $N_c(d+1) = N_c(d) + N_{c-1}(d)$, so $N_c(d+1) - N_{c+1}(d+1) = N_{c-1}(d) - N_{c+1}(d)$, and by induction the three counts differ pairwise by at most one. Apply Theorem 3.14 with $n = 2^d$, using $2^d \equiv 1$ or $2 \pmod 3$ according as d is even or odd. □

This gives $2, 6, 12, 30$ for $d = 2, 3, 4, 5$. Tori behave the same way: for example, colouring $(i,j) \mapsto i + j \bmod 3$ shows that $\gamma(C_{3a} \square C_{3b}, K_3) = 6ab$ attains the pigeonhole bound.

3.3 Joins with an independent set

The *join* $G_1 + \overline{K_k}$ is obtained from G_1 by adding k pairwise non-adjacent vertices, each joined to every vertex of G_1. For a bipartite G_1, let $\beta(G_1)$ be the least possible size of the larger class of a proper 2-colouring of G_1. Put $\beta(G_1) = \infty$ if G_1 is not bipartite.

Theorem 3.17. *Let G_1 have at least one edge and let $k \geq 1$. Then*

$$\gamma(G_1 + \overline{K_k}, K_3) = \max\bigl(|E(G_1)|,\ k\,\beta(G_1)\bigr).$$

Proof. Let $f : G_1 + \overline{K_k} \to K_3$, and let c be the colour of one new vertex. No vertex of G_1 has colour c. If another new vertex had a colour $c' \neq c$, the two ends of an edge of G_1 would need two distinct colours outside $\{c, c'\}$, which is impossible. So all new vertices have colour c, and f restricts to a 2-colouring of G_1 with classes A and B in the other colours a, b. Every edge of G_1 maps onto ab, and the edges at the new vertices give loads $k|A|$ on ac and $k|B|$ on bc. Hence $\mu(f) = \max(|E(G_1)|, k|A|, k|B|)$. Minimising over 2-colourings of G_1 gives the formula. □

Corollary 3.18. *$\gamma(B_k, K_3) = k$, $\gamma(F_k, K_3) = k$ and $\gamma(W_n, K_3) = n - 1$ for odd $n \geq 5$. Also $\gamma(K_{p,q,r}, K_3) = \max(pq, r\max(p,q))$ for $p, q, r \geq 1$, which equals $\max(pq, pr, qr)$. For the fan $P_t + K_1$ with $t \geq 3$, $\gamma = t - 1$.*

Proof. Take $G_1 = K_2$ and k new vertices for B_k. Take $G_1 = kK_2$ or C_{n-1} with one new vertex for F_k and W_n. Take $G_1 = K_{p,q}$ for $K_{p,q,r}$, and $G_1 = P_t$ for the fan. □

3.4 Irregular source graphs

Theorem 3.19 gives a direct proof of the first four cases of Corollary 3.18, with the values of $|\cdot, K_2|$ as well.

The *book* B_k is k triangles sharing one edge (the spine). The *friendship graph* F_k is k triangles sharing one vertex. The *wheel* W_n is a vertex (the hub) joined to every vertex of C_{n-1}.

Theorem 3.19. *1. For $k \geq 1$: $\gamma(B_k, K_3) = k$ and $|B_k, K_2| = 1$.*

2. For $k \geq 0$: $\gamma(F_k, K_3) = k$ and $|F_k, K_2| = k$.

3. For odd $n \geq 5$: $\gamma(W_n, K_3) = n - 1$.

4. For $p, q, r \geq 1$: $\gamma(K_{p,q,r}, K_3) = \max(pq, pr, qr)$.

Proof. All four use Lemma 3.9.

(1) The two spine vertices get distinct colours a, b, and every page vertex gets the third colour c. So $D_a = D_b = k + 1$ and $D_c = 2k$, giving $L_{ab} = 1$ and $L_{ac} = L_{bc} = k$. Deleting the spine leaves a bipartite graph, and B_k contains a triangle, so $|B_k, K_2| = 1$.

(2) The hub gets some colour c. Each triangle puts one vertex in each of the other two classes, so $D_0 = D_1 = D_2 = 2k$ and every load equals k, for every homomorphism. The k triangles are edge-disjoint, so $|F_k, K_2| \geq k$. Deleting the k edges opposite the hub leaves a star.

(3) The hub, of colour c, is alone in its class, so $D_c = n - 1$. The total degree is $4(n-1)$, so $D_a + D_b = 3(n-1)$ and $L_{ab} = n - 1$. Alternating two colours on the even rim attains this.

(4) If all parts are non-empty, each part is monochromatic and the three parts get distinct colours. Otherwise some vertex would see three colours among its neighbours. The class of part i has $D = s_i(N - s_i)$, where $N = p+q+r$, so $2L = s_i(N-s_i) + s_j(N-s_j) - s_k(N-s_k) = 2s_is_j$. □

In particular $\gamma(F_k, K_3)$ attains the pigeonhole bound k. On the other hand $\gamma(W_n, K_3) = n-1$ exceeds it for $n \geq 5$, and so do $\gamma(K_{1,2,3}, K_3) = 6 > 4$ and $\gamma(K_{2,3,4}, K_3) = 12 > 9$. The book graphs separate the two parameters: $|B_k, K_2| = 1$ for every k, while $\gamma(B_k, K_3) = k$ grows without bound.

3.5 Cycles into odd cycles

Theorem 3.20. *Let $m \geq 3$ be odd and $n \geq 3$, and suppose $C_n \to C_m$. Then $\gamma(C_n, C_m)$ is the least integer M such that $M \equiv n \pmod 2$ and $mM \geq n$. Thus it is $\lceil n/m \rceil$ or $\lceil n/m \rceil + 1$, whichever has the parity of n.*

Proof. Fix an edge $e = \{x, x+1\}$ of C_m. Number the vertices of the path $C_m - e$ from $x+1$ to x as $0, 1, \ldots, m-1$. Along the closed walk $f(0), f(1), \ldots$ the position changes parity at every step except a step across e, where it goes between 0 and the even number $m-1$. A closed walk changes parity an even number of times, so $n - \mu(f, e)$ is even. Hence every load, and in particular $\mu(f)$, has the parity of n. The pigeonhole bound gives $m\mu(f) \geq n$.

For the upper bound write $n = wm + 2s$ with $w \in \{0, 1\}$ the parity of n. Walk once around C_m if $w = 1$. Then walk s steps forward and s steps back. Each edge is used at most $w + 2\lceil s/m \rceil \leq M$ times. □

For $m = 3$, where $C_3 = K_3$, this is Theorem 3.11. The smallest M of the parity of n with $3M \geq n$ is $\lceil n/3 \rceil$, except that it is $\lceil n/3 \rceil + 1$ when $n \equiv 2 \pmod 3$. For $m = 5$ and $m = 7$ it gives, for example, $\gamma(C_6, C_5) = 2$, $\gamma(C_{12}, C_5) = 4$, $\gamma(C_9, C_7) = 3$ and $\gamma(C_{15}, C_7) = 3$. So the pigeonhole bound $\lceil n/m \rceil$ is attained exactly when $\lceil n/m \rceil \equiv n \pmod 2$.

The parity phenomenon extends from cycles to all regular graphs.

Theorem 3.21. *Let G be d-regular on n vertices, $d \geq 1$, and let $m \geq 3$ be odd. For $f\colon G \to C_m$ put $n_c = |f^{-1}(c)|$ and $s_c = \sum_{t=0}^{m-1} (-1)^t n_{c-t}$. Then the load of the edge $\{c, c+1\}$ is $\frac{d}{2} s_c$. Consequently*

$$\gamma(G, C_m) \;\geq\; \frac{d}{2} M_m(n),$$

where $M_m(n)$ is the least $M \equiv n \pmod 2$ with $mM \geq n$. Equality holds whenever some f has $s_c \leq M_m(n)$ for all c.

Proof. Write L_c for the load of $\{c, c+1\}$. Every edge at a vertex of colour c goes to colour $c \pm 1$, so $d\, n_c = L_{c-1} + L_c$. Telescoping this relation once around the odd cycle gives $2L_c = \sum_t (-1)^t d\, n_{c-t} = d\, s_c$. Hence $s_c \geq 0$, $s_c + s_{c-1} = 2n_c$, so all s_c have the same parity, and $\sum_c s_c = \frac{2}{d}|E(G)| = n$. The largest s_c therefore satisfies $s_c \equiv n \pmod 2$ and $m\, s_c \geq n$, so it is at least $M_m(n)$. □

For $m = 3$ this is Theorem 3.10, since s_c is then $a + b - c$ in the notation used there. For $d = 2$ it is the lower bound of Theorem 3.20. A computer search finds the bound exact for the tori $C_5 \square C_5$, $C_5 \square C_7$ and $C_7 \square C_7$ into C_5 and $C_7 \square C_7$ into C_7. It is also exact for 412 of the 465 triangle-free connected cubic graphs on 10 to 16 vertices that map to C_5; the other 53 exceed it by exactly 3.

Theorem 3.22. *Let $m \geq 3$ be odd and $n \geq 3$. Then $C_n \to C_m$ if and only if n is even or $n \geq m$.*

Proof. An even cycle maps onto an edge. If $n \geq m$ are both odd, wind once around C_m and then oscillate on an edge for the remaining even number of steps. Conversely, let $n < m$ with n odd and $f : C_n \to C_m$. Then f misses some vertex w, and $C_m - w$ is a path. So C_n would be bipartite, a contradiction. □

4 The Symmetric Combination $m(G, H)$

Let $m(G, H) = \max(|G, H|, |H, G|)$. It is non-negative and symmetric, and $m(G, G) = 0$.

Proposition 4.1. *$m(G, H) = 0$ if and only if $G \leftrightarrow H$.*

Theorem 4.2. *m does not satisfy the triangle inequality, and so is not a pseudo-metric.*

Proof. By Theorem 2.4, and since $K_n \to K_m$ for $n \leq m$,

$$m(K_7, K_3) = 5, \qquad m(K_7, K_5) = 2, \qquad m(K_5, K_3) = 2,$$

and $5 > 2 + 2$. □

Remark 4.3. Khoshkha [1] claims that m is a pseudo-metric. Theorem 4.2 shows that this is false under Definition 2.1.

5 Applications

The exact values in this paper rarely beat the obvious designs by much. Their main use is as *certified ground truth*: optima at any size, proved in Lean for families of graphs and targets that people actually use.

Benchmarks for exact and heuristic solvers. Max-k-Cut, minimum-violation H-colouring, QUBO and Ising solvers, including quantum annealers, are usually evaluated against best-known values or planted solutions. The families above give instances with proved optima, and one instance serves every target with the same clique number. The informative ones are those where any relaxation or averaging argument stops short of the optimum, so that a solver has to find a global proof:

Instance	optimum	naive bound
$\lvert L(K_7), K_2 \rvert$	43	42
$\gamma(Q_{10}, K_3)$	1710	1707
$\gamma(Q_{11}, K_3)$	3762	3755
$\gamma(C_7 \square C_7, C_5)$	22	20

Corollary 3.15 adds an unlimited supply of irregular instances: every random connected cubic graph other than K_4 and $K_{3,3}$ has $\gamma(G, K_3) = 3\lceil |V(G)|/6 \rceil$. This rests on the cited Chen–Lih–Wu theorem.

Frustrated magnets. Corollary 2.15 gives exact ground-state energies on kagome, checkerboard and pyrochlore tori for every target. The products $C_p \times C_q$ with $p \leq q$ odd carry a forced seam of exactly $2p$ unsatisfied bonds for every target into which C_q does not map (Theorem 2.22). These are natural test lattices for annealing and Monte Carlo methods. The theorems say nothing about ground-state degeneracy or entropy.

Placement across failure domains. Spreading the switches of a network over m failure domains, so that as few links as possible stay inside a domain, is exactly the problem $|G, K_m|$. On HyperX and flattened-butterfly fabrics, which are Hamming graphs, Theorem 2.7 gives the optimum. For example $H(3, 8)$ split into four domains keeps exactly 768 of its 5376 links inside domains. For folded cubes the optimum 2^{n-2} of Theorem 2.11 is attained by removing the antipodal links. In a d-regular network whose links must all cross domains, Theorem 3.10 shows that the load between two domains depends only on the domain sizes.

These applications do not cover locality-preserving partitioning, which minimises the links between domains rather than inside them, or access patterns such as strides and diagonals in memory-bank layouts.

6 Formal Verification

The Lean 4 development in `lean/NonHom` uses Lean and Mathlib version 4.35.0. It consists of 33 files and about 8 000 lines. It defines $|G, H|$ as the infimum, in $\mathbb{N} \cup \{\infty\}$, of $|S|$ over finite edge sets S such that $G - S \to H$. It defines $\gamma(G, H)$ as the infimum of $\mu(f)$ over all homomorphisms, which is ∞ when there are none.

There is no `sorry` and no `native_decide`. Every theorem depends only on Lean's standard axioms (`propext`, `Classical.choice`, `Quot.sound`). Finite checks, such as the facts about the 5-cycles and perfect matchings of P and the explicit colourings, are done with `decide +kernel`, which the Lean kernel checks. Turán's theorem and Mantel's theorem are taken from Mathlib. Table 2 lists the Lean name of each result.

Two parts of the paper are outside the formal development. Theorem 2.23 is formalized for arbitrary disjoint families of sets, each left by exactly one edge; that leaf blocks form such a family is not formalized. In Theorem 2.12 the equality is formalized given a balanced colouring, and in particular for cubic graphs with a perfect matching. The classical existence results of Remark 2.13 are not formalized. Corollary 3.15 combines the formalized Theorem 3.14 with the Chen–Lih–Wu theorem, which is not formalized. Corollary 2.15 combines Theorem 2.14 with Kőnig's theorem, which is not formalized. The computer searches after Theorem 3.21 are not formalized. The exhaustive search for $L(K_7)$ and the SAT computation for $L(K_{11})$ after Theorem 2.17 are not formalized. The torus example after Theorem 3.16 is an unformalized remark.

7 Open Questions

1. **General targets.** Lemma 3.9 determines $\mu(f)$ for $f\colon G \to K_3$ from the degree sums of the colour classes. Is there an analogous identity for targets H other than K_3?

2. **When is the pigeonhole bound attained?** For cycles into K_3 it is attained exactly when $n \not\equiv 2 \pmod 3$ (Theorem 3.11). For d-regular G, Theorem 3.10 turns the question into one about realisable colour-class sizes. Characterise the pairs (G, H) for which $\gamma(G, H) = \lceil |E(G)|/|E(H)| \rceil$.

3. **Tightness of the transitive bound.** Theorem 2.10 is tight for the Petersen graph and for $C_p \times C_q$. For which edge-transitive G is $|G, K_2|$ equal to $|E(G)|$ divided by the odd girth?

4. **Joins with other targets.** Is there a formula analogous to Theorem 3.17 for $\gamma(G_1 + \overline{K_k}, K_m)$ with $m \geq 4$, or for general joins $G_1 + G_2$?

5. **Triangular graphs.** Is $|L(K_n), H| = |L(K_n), K_2|$ for every triangle-free H with an edge? By Theorem 2.12 this holds for $n \not\equiv 3 \pmod 4$, and the computations after Theorem 2.17 confirm it for $n = 7$ and $n = 11$. It would follow from this statement: for balanced local bipartitions of K_n, if no triple has three separators then no triple has one. For $r \geq 3$ the answer is negative by Theorem 2.18.

6. **Repairing m.** Does the edge-edit distance to a graph mapping to H, or m restricted to graphs with the same number of vertices, satisfy the triangle inequality?

7. **Petersen colourings.** Is $|L(G), L(P)|$ equal to the number of leaf blocks of the bridge-block tree for every cubic G? The case of bridgeless G is Jaeger's conjecture.

8. **Tightness of Theorem 2.21.** For which G, H is the bound tight for $K = K_2$ or $K = K_3$? Theorem 2.22 gives an infinite tight family for K_2. For large targets the bound fails to be tight, by [5].

Table 2: Lean names of the results.

Result	Lean theorem
Props. 1.2, 1.3	`chromaticNumber_le_of_hom`, `edist_map_le`
Prop. 2.3	`nhf_le_of_hom_target`, `nhf_eq_of_hom_equiv`, `nhf_le_of_edge_injective`
Thm. 2.4, Table 1	`nhf_top_top_formula`, `nhf_top_succ_top`, `table_KnKm`
Thm. 2.5	`lower_bound`
Thm. 2.6	`nhf_top_clique`, `nhf_top_triangleFree`
Thm. 2.8	`nhf_average`
Thm. 2.10	`nhf_edge_transitive`
Thm. 2.11	`nhf_folded`, `nhf_folded_K2`, `nhf_folded_cycle`
$\lvert C_n, K_3\rvert$, $\lvert C_n, K_2\rvert$	`nhf_cycle_K3`, `nhf_cycle_K2_even`, `nhf_cycle_K2_odd`
Thm. 2.9	`nhf_petersen_K2`
Thm. 2.12	`line_lower_clique`, `line_eq`, `line_cubic_pm`
Thm. 2.14	`nhf_line_class1`
Thm. 2.16	`sq_lower`
Thms. 2.21, 2.22	`hedetniemi_bound`, `nhf_tensor_cycles_target`, `nhf_tensor_cycles_zero`, `nhf_tensor_cycles_cycle`
Thm. 2.23	`jaeger_lower`
Prop. 3.2, Lemma 3.3	`ceil_div_le_gamma`, `one_le_gamma`, `gamma_self`
Lemmas 3.4, 3.5, 3.7	`gamma_comp_le`, `nhf_le_gamma_mul`, `gamma_le_of_injective_hom`
Cor. 3.6, Prop. 3.8	`two_sided`, `gamma_top_top_eq_one`, `gamma_top_top_eq_top`
Lemma 3.9	`two_load_eq`, `two_mu_eq_deg`
Thm. 3.10	`two_mu_eq`, `gamma_K3_lower`, `gamma_K3_upper`
Thms. 3.11, 3.12	`gamma_cycle`, `gamma_knn`, `nhf_knn_K3`
Thm. 3.13	`gamma_petersen_K3`, `gamma_petersen_K5`
Thm. 2.7	`nhf_ham`
Thm. 2.17	`nhf_triangular`
Thm. 2.19	`tri_lower_parity`, `nhf_triangular_even`, `nhf_triangular_equitable`, `nhf_triangular_odd_div_even`, `nhf_triangular_odd_div_odd`
Thm. 2.18	`nhf_triangular_general_target`
Cor. 2.20	`cutSize_le_of_lapForm_le`, `isLeast_triangular`, `isLeast_ham`, `exists_cutSize_triangular_eq_iff`, `exists_cutSize_ham_eq_iff`
Thm. 3.14	`gamma_K3_equitable`
Thm. 3.16	`gamma_cube`, `Mtri_two_pow`
Thm. 3.17	`gamma_join`
Thm. 3.20	`gamma_cycle_cycle`
Thm. 3.21	`two_Lc`, `gamma_oddcycle_lower`, `gamma_oddcycle_upper`
Thm. 3.19	`gamma_book`, `nhf_book_K2`, `gamma_friendship`, `nhf_friendship_K2`, `gamma_wheel`, `gamma_tri`
Thm. 3.22 and examples	`cycle_hom_iff`, `gamma_C6_C5`, `gamma_C15_C7`, ...
Prop. 4.1, Thm. 4.2	`msym_eq_zero_iff`, `no_triangle`, `msym_not_triangle`

References

References

[1] K. Khoshkha, *Nonhomomorphism Factors*, M.Sc. thesis, Sharif University of Technology, 2005.

[2] A. Daneshgar and H. Hajiabolhassan, "Circular colouring and algebraic no-homomorphism theorems," *European Journal of Combinatorics*, vol. 28, no. 6, pp. 1843–1853, 2007.

[3] P. Hell and J. Nešetřil, *Graphs and Homomorphisms*, Oxford University Press, 2004.

[4] F. Foucaud, H. Hocquard, D. Lajou, V. Mitsou and T. Pierron, "Parameterized complexity of edge-coloured and signed graph homomorphism problems," in *14th International Symposium on Parameterized and Exact Computation (IPEC 2019)*, LIPIcs vol. 148, pp. 15:1–15:16, 2019.

[5] Y. Shitov, "Counterexamples to Hedetniemi's conjecture," *Annals of Mathematics*, vol. 190, no. 2, pp. 663–667, 2019.

[6] J. Petersen, "Die Theorie der regulären graphs," *Acta Mathematica*, vol. 15, pp. 193–220, 1891.

[7] D. Kőnig, "Über Graphen und ihre Anwendung auf Determinantentheorie und Mengenlehre," *Mathematische Annalen*, vol. 77, pp. 453–465, 1916.

[8] F. Jaeger, "Nowhere-zero flow problems," in *Selected Topics in Graph Theory 3*, Academic Press, pp. 71–95, 1988.

[9] L. de Moura and S. Ullrich, "The Lean 4 theorem prover and programming language," in *Automated Deduction – CADE 28*, LNCS vol. 12699, pp. 625–635, 2021.

[10] B.-L. Chen, K.-W. Lih and P.-L. Wu, "Equitable coloring and the maximum degree," *European Journal of Combinatorics*, vol. 15, pp. 443–447, 1994.

[11] T. Färnqvist, P. Jonsson and J. Thapper, "Approximability distance in the space of H-colourability problems," in *Computer Science – Theory and Applications (CSR 2009)*, 2009. arXiv:0802.0423.

[12] R. Engström, T. Färnqvist, P. Jonsson and J. Thapper, "Graph homomorphisms, circular colouring, and fractional covering by H-cuts," arXiv:0904.4600, 2009.

[13] B. Mohar and S. Poljak, "Eigenvalues and the max-cut problem," *Czechoslovak Mathematical Journal*, vol. 40, pp. 343–352, 1990.

[14] E. R. van Dam and R. Sotirov, "New bounds for the max-k-cut and chromatic number of a graph," *Linear Algebra and its Applications*, vol. 488, pp. 216–234, 2016.

[15] A. J. W. Hilton and D. de Werra, "A sufficient condition for equitable edge-colourings of simple graphs," *Discrete Mathematics*, vol. 128, 1994.

[16] É. Lucas, *Récréations mathématiques*, vol. 2, Gauthier-Villars, 1892 (Walecki's construction).

[17] The mathlib Community, "The Lean mathematical library," in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pp. 367–381, 2020.